自动驾驶决策规划算法第二章第6节(中)

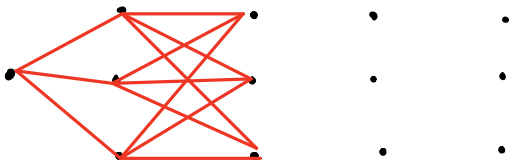
算力已经捉襟见肘

重体验，轻细节，学会以后请立即转 C++

未来：核心功能 C++

画图，可视化工具 python 学会算法以后 Matlab直接废弃

背景：dp + qp



dp

$$s_{i+1} = s_i + \dot{s}_i\,dt + \frac{1}{2}\ddot{s}_i\,dt^2 + \frac{1}{6}\left(\frac{\dddot{s}_{i+1} - \dddot{s}_i}{dt}\right)dt^3$$

$$\dot{s}_{i+1} = \dot{s}_i + \ddot{s}_i\,dt + \frac{1}{2}\left(\frac{\dddot{s}_{i+1} - \dddot{s}_i}{dt}\right)dt^2$$

qp

算法步骤

第一部分：生成ST图

开始：已有路径规划的结果 trajectory x, y, theta, kappa

第一步，生成 path 的 index2s

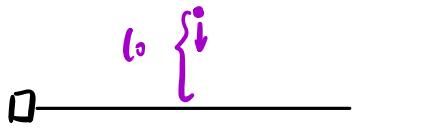
```
for i = 2 : length(trajectory.x)
    index2s(i) = index2s(i-1) + sqrt(dx^2 + dy^2)
end
```

第二步，将动态障碍物投影到以路径规划path为坐标轴的 frenet坐标系上

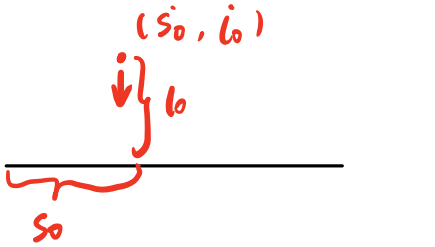


得到动态障碍物的坐标 $(s_0, l_0, \dot{s}_0, \dot{l}_0)$

第三步，动态障碍物投影，生成ST图



预测 dynamic obs 在 frenet 坐标做匀速运动



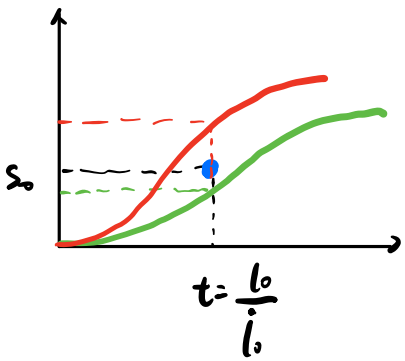
问，红点什么时候到达path

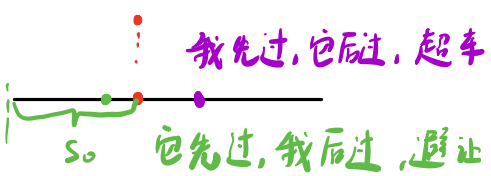
答：$t = \frac{l_0}{\dot{l}_0}$

问：红点切入，切出时的纵向s是什么

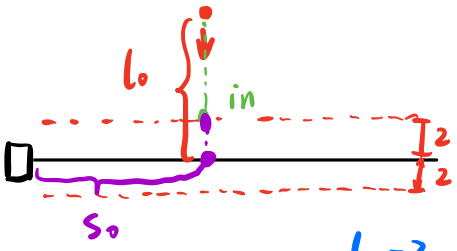
答：切入 s_0，切出也是 s_0

画ST图





精微复杂一些



切入时间 $t_in = \frac{l_0 - 2}{i_0} = \frac{l_0}{i_0} + \frac{-2}{i_0}$

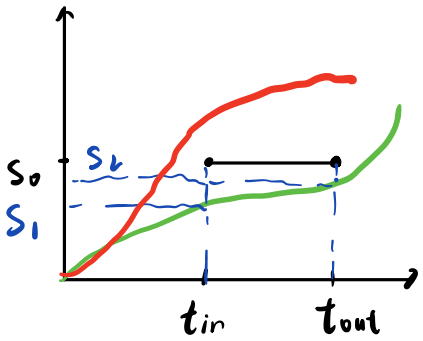
$\frac{l_0}{i_0}$：到达 path 的时间

$\frac{-2}{i_0}$：在前时光倒流 -2m 所需要的时间

切出时间 $t_out = \frac{l_0}{i_0} + \frac{2}{i_0}$

$s_in,\ s_out = s_0$

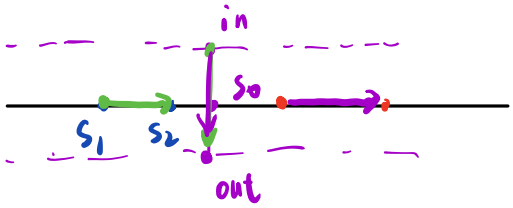
ST 图



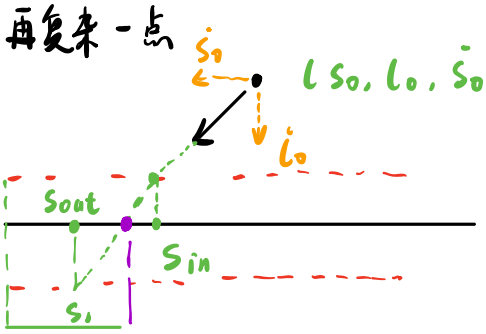
超车

在 (t_in, t_out) 区间内，host 的 $s < s_0$

避让



再复杂一点



到达 path 的时间 $t = l_0 / i_0$

到达 path 的纵向距离 $s_1 = s_0 + \dot{s}_0 t = s_0 + \dot{s}_0 \cdot \frac{l_0}{i_0}$

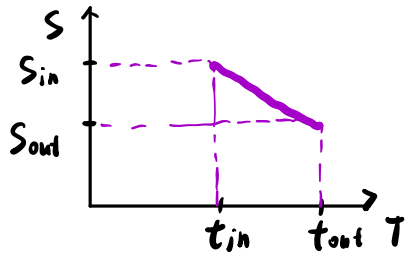
$t_{in}, t_{out} = t \pm \frac{2}{i_0}$ t_{in} 永远小于 t_{out}，可以先算 $t + \frac{2}{i_0}$ $t - \frac{2}{i_0}$，再比大小

把小的给 t_{in}，大的给 t_{out}，省去正负号判断的困扰

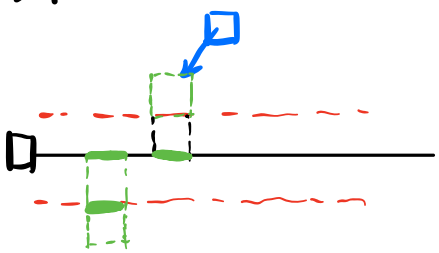
计算 s_{in}, s_{out} $s_{in} = s_0 + s_0 t_{in}$ 到达上边界时，S 走过的距离

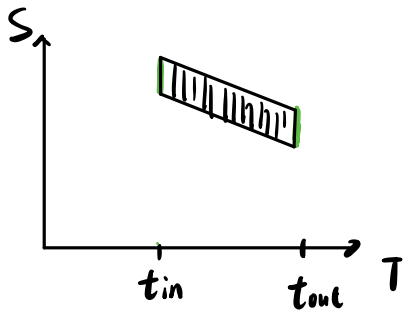
$s_{out} = s_0 + s_0 t_{out}$

画图

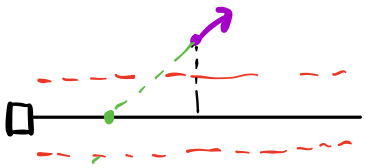


再复杂一点





特殊 case



$t = \frac{0 - i_0}{i} < 0$ $t_{in/out} < 0$ 忽略

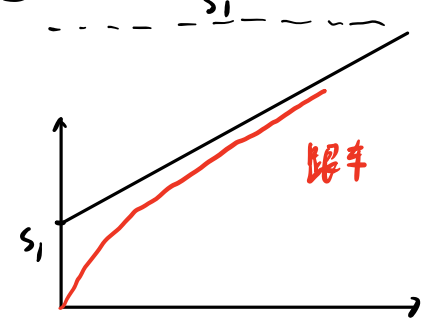
$t_{in/out} = t \pm \frac{2}{i_0}$

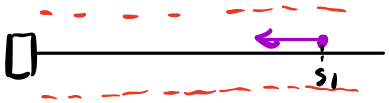


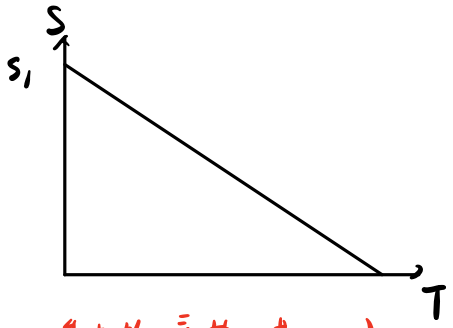
$t_{in} = 0$ $t_{out} > 0$



$t_{in} = 0$, $t_{out} = +\infty$

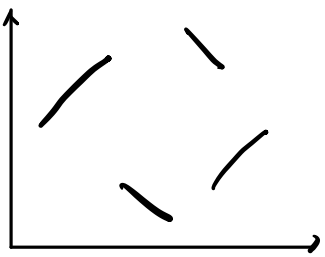






纯靠速度规划无解 (具体做法参考第二章第七节(上))

本教程的 ST 图以斜直线的模型展示



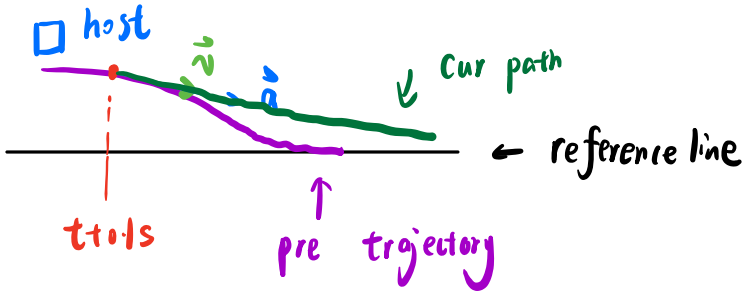
只考虑 host 的碰撞体积，不考虑 obs 的体积

为什么？ 第九问题



第二部分 dp + qp

第一步：确定规划起点

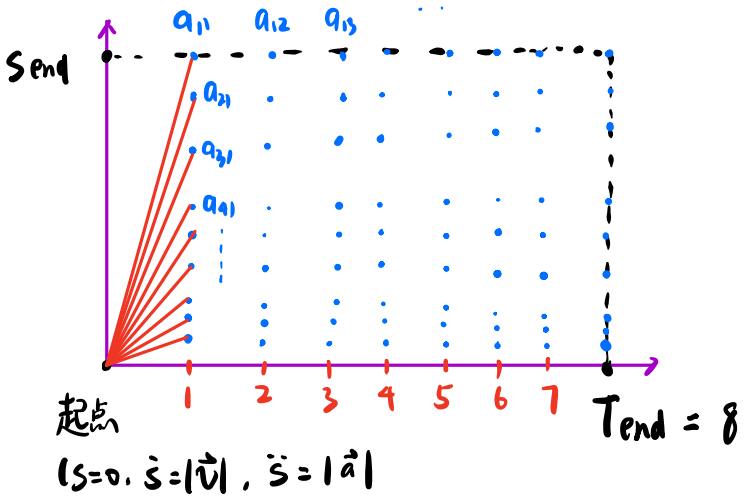


$(x, y, \theta, k, \vec{v}, \vec{a})$ $\vec{a}$ 为切向加速度

无论 cur path 的形状如何，cur path 的起点的切线方向与拼接点保持一致

∴ 速度规划的起点，$s=0$，$\dot{s}=|\vec{v}|$，$\ddot{s}=|\vec{a}|$

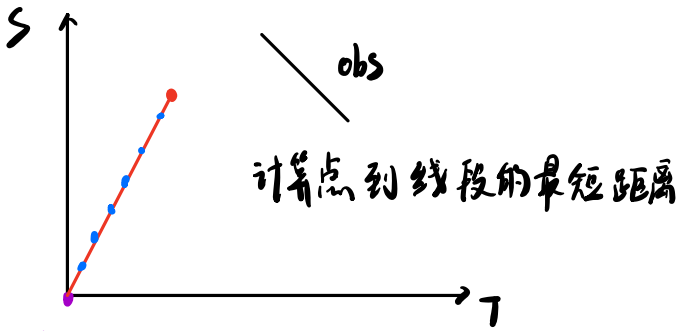
第二步：动态规划



Send是路径规划的path的长度

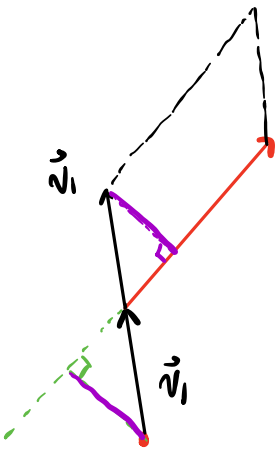
$(S=0, \dot{S}=|\vec{v}|, \ddot{S}=|\vec{a}|)$

dp-cost = 障碍物代价 + 加速度代价 + 推荐速度代价 + jerk 代价



计算点到线段的最短距离

起点 $(S=0, T=0, \dot{S}=|\vec{v}|, \ddot{S}=|\vec{a}|)$



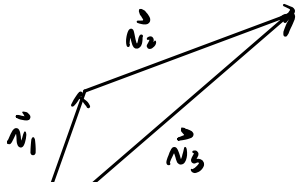
$$高 = \frac{面积}{|\vec{v}_2|}$$

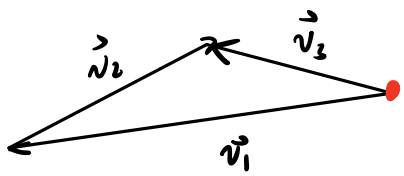
面积 = $\vec{v}_1 \times \vec{v}_2$ (向量积)

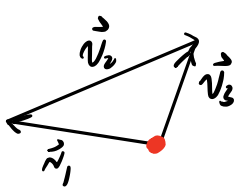
$$d_1 = |\vec{v}_1|$$

$$d_2 = |\vec{v}_1 + \vec{v}_2|$$

$$d_3 = \frac{|\vec{v}_1 \times \vec{v}_2|}{|\vec{v}_2|}$$







$\vec{v}_1 \cdot \vec{v}_2 > 0$　　$\vec{v}_1 \cdot \vec{v}_2 < 0$　　$\vec{v}_1 \cdot \vec{v}_2 < 0$

$\vec{v}_3 \cdot \vec{v}_2 > 0$　　$\vec{v}_3 \cdot \vec{v}_2 < 0$　　$\vec{v}_3 \cdot \vec{v}_2 > 0$

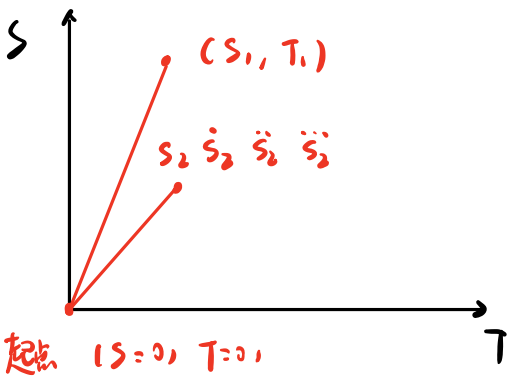
若 $\vec{v}_1 \cdot \vec{v}_2$ 与 $\vec{v}_3 \cdot \vec{v}_2$ 同号

$min_d = min(|\vec{v}_1|, |\vec{v}_3|)$

若 ——— 异号

$$min_d = \frac{|\vec{v}_1 \times \vec{v}_2|}{|\vec{v}_2|}$$

$$cost_obs = \begin{cases} \infty & |min_d| < 2 \\ 衰减函数 & 2 < |min_d| < 3 \\ 0 & |min_d| > 3 \end{cases}$$



起点 $(S=0, T=0)$

$\dot{S} = |\vec{v}|$　$\ddot{S} = |\vec{a}|$

$$\dot{S}_1 = \frac{S_1 - 0}{T_1 - 0}$$

$$\ddot{S}_1 = \frac{\dot{S}_1 - |\vec{v}|}{T_1 - 0}$$

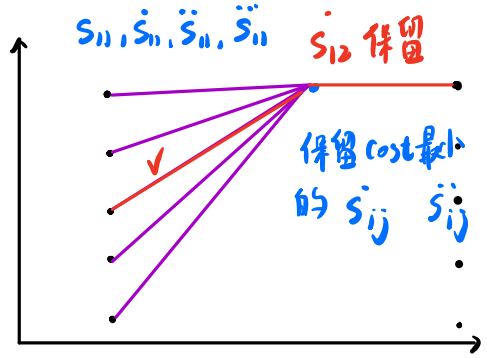
$$\dddot{S}_1 = \frac{\ddot{S}_1 - |\vec{a}|}{T_1 - 0}$$

$cost_ref_speed = W_{ref_speed} \cdot (\dot{S}_1 - v_r)^2$

v_r 由地图决定

$cost_accel = W_{cost_accel} (\ddot{S}_1)^2$

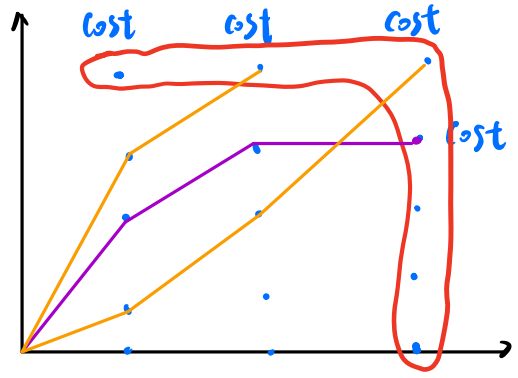
$cost_jerk = W_{cost_jerk} (\dddot{S}_1)^2$



$$\dot{S}_{13} = \frac{S_{13} - S_{12}}{T_{13} - T_{12}}$$

$$\ddot{S}_{13} = \frac{\dot{S}_{13} - \dot{S}_{12}}{T_{13} - T_{12}}$$

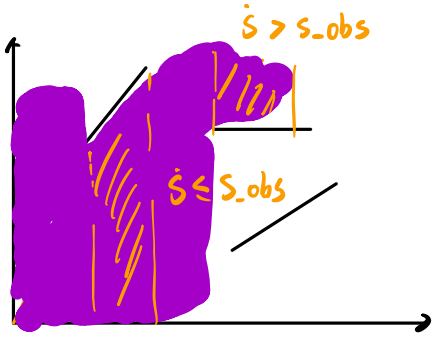
动态规划具体算法见路径规划篇



遍历该区域，找到cost最小的作为路径规划的终点

递推，得到 dp_speed_s，dp_speed_t

第三步：二次规划



生成凸空间 生成 s_lb，s_ub，ṡ_lb，ṡ_ub

有3 path，也就有3 kappa，以 $a_{ymax} = 0.2 \times 9.8$ $\therefore \frac{v^2}{R} \le a_{ymax}$

$\Rightarrow v^2 \le R \cdot a_{ymax} = \frac{a_{ymax}}{kappa}$ $0 \le \dot{s} \le \left(\frac{a_{ymax}}{kappa}\right)^{\frac{1}{2}}$

动力学约束 $-6 \le \ddot{s} \le 4$

不允许倒车约束 $s_{i+1} \ge s_i$

连续性约束 $s_{i+1} = s_i + \dot{s}_i dt + \frac{1}{2}\ddot{s}_i dt^2 + \frac{1}{6}\left(\frac{\ddot{s}_{i+1} - \ddot{s}_i}{dt}\right)dt^3$

$\dot{s}_{i+1} = \dot{s}_i + \ddot{s}_i dt + \frac{1}{2}\left(\frac{\ddot{s}_{i+1} - \ddot{s}_i}{dt}\right)dt^2$

$cost_function = \sum w_1(s_i - v_{ref})^2 + w_2(\dot{s}_i)^2 + w_3(\ddot{s}_i)^2$

第三部分，后处理

path & speed 融合成 trajectory

增密 trajectory，坐标转换

拼接轨迹